Dimension Reduction with Latent Variable Models for Simplification of Image Data

Joey Hotz

Department of Statistical Sciences, University of Toronto

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Presentation Agenda

1. Introduction
2. Methods for Dimension Reduction
3. Data Description
4. Results
5. Discussion

Motivations & Research Goals

Primary Goal

Investigate properties of **Probabilistic Principal Component Analysis (PPCA)** and **Gaussian Process Latent Variable Models (GP-LVM)**

PPCA (Tipping and Bishop, 1999) and GP-LVM (Lawrence, 2005) are both extensions of principal component analysis which search for latent variable representations of high-dimensional data.

Research Question

Research Question

Which latent variable methods are best suited for reducing the dimensionality of image data?

Performance metrics:

- Cluster–Label Agreement between clusters in latent variable space and true class labels.
- Reconstruction Error from mapping latent variables back to original image data.

Setting: Dimension Reduction

Suppose we have some feature matrix $\mathbf{Y} \in \mathbb{R}^{N \times D}$, containing N rows of observations $\mathbf{y}_i^\mathsf{T} \in \mathbb{R}^D$.

To avoid the *Curse of Dimensionality*, employ a mapping $\phi : \mathbb{R}^D \to \mathbb{R}^Q$ and work with lower-dimensional data $\phi(\mathbf{y}_i) \in \mathbb{R}^Q$.

Methods for Dimension Reduction

1: Principal Component Analysis (PCA)

2: Probabilistic Principal Component Analysis (PPCA)

3: Gaussian Process Latent Variable Models (GP-LVM)

Principal Component Analysis

PCA projects each observation $\mathbf{y}_i \in \mathbb{R}^D$ onto an Q-dimensional hyperplane which minimizes reconstruction error.

The principal components are the top Q eigenvectors from the empirical covariance matrix $\mathbf{S} \in \mathbb{R}^{D \times D}$ for the feature matrix $\mathbf{Y}$.

Principal Component Analysis

Benefits of PCA

- Nice mathematical closed-form representation and interpretation.
- Relatively quick to compute.

Downsides of PCA

- PCA enforces a *linear* relationship between observations and lower-dimensional subspace.
- Lacks a direct probabilistic interpretation.

Probabilistic Principal Component Analysis

PPCA is similar in nature to 'standard' PCA with added distributional assumptions.

PPCA Model: Each observation $\mathbf{y}_i$ is generated by latent variables $\mathbf{u}_i$ according to an *affine* linear mapping

$$\mathbf{y}_i = \mathbf{W}\mathbf{u}_i + \boldsymbol{\mu} + \boldsymbol{\epsilon}_i$$

Elements of the PPCA Distribution

- $\mathbf{W} \in \mathbb{R}^{D \times Q}$
- $\boldsymbol{\mu} \in \mathbb{R}^{D}$
- $\mathbf{u}_i \in \mathbb{R}^{Q}$
- $\boldsymbol{\epsilon}_i \in \mathbb{R}^{D}$, with $\boldsymbol{\epsilon}_i \sim \mathcal{N}_D(\mathbf{0}, \boldsymbol{\Psi})$

Probabilistic Principal Component Analysis

When $\mathbf{u}_i \sim \mathcal{N}_Q(\mathbf{0}, \mathbf{I}_Q)$ and $\text{Cov}(\boldsymbol{\epsilon}_i) = \boldsymbol{\Psi} = \sigma^2 \mathbf{I}_D$, we have that

$$\mathbf{y}_i | \mathbf{u}_i \sim \mathcal{N}_D\big(\mathbf{W}\mathbf{u}_i + \boldsymbol{\mu}, \sigma^2 \mathbf{I}_D\big)$$

Integrating the conditional distribution above with respect to $\mathbf{u}_i$ yields the marginal distribution for the observations $\mathbf{y}_i \in \mathbb{R}^Q$

$$\mathbf{y}_i \sim \mathcal{N}_D\big(\boldsymbol{\mu}, \mathbf{W}\mathbf{W}^\mathsf{T} + \sigma^2 \mathbf{I}_D\big)$$

Probabilistic Principal Component Analysis

The distributional form of $\mathbf{y}_i$ implies maximum likelihood estimators for elements of the PPCA model:

$$\boldsymbol{\mu}_{\text{MLE}} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{y}_i$$

$$\mathbf{S}_{\text{MLE}} = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{y}_i - \boldsymbol{\mu}_{MLE})(\mathbf{y}_i - \boldsymbol{\mu}_{MLE})^{\mathsf{T}}$$

Under the PPCA modelling assumptions, we also can derive *exact* closed-form equations for the MLE of $\mathbf{W}$ and σ^2:

$$\mathbf{W}_{\text{MLE}} = \mathbf{U}_{(Q)}\left(\boldsymbol{\Lambda}_{(Q)} - \sigma^2\right)^{1/2}\mathbf{R}$$

$$\sigma^2_{\text{MLE}} = \frac{1}{D-Q}\sum_{j=Q+1}^{D}\lambda_j$$

Probabilistic Principal Component Analysis

Benefits of PPCA

- Closed-form equations for maximum likelihood estimators.
- Convenient probabilistic interpretation.
- Relatively quick to compute.

Downsides of PPCA

- PPCA enforces a *linear* relationship between the expected observations and the latent variables.

Gaussian Processes - Introduction

A *Gaussian Process* $\{X_t\}_{t \in T}$ is a stochastic process where any finite collection of variables follows a joint multivariate Gaussian distribution.

Gaussian Process Specification

A Gaussian Process is fully specified by functions $\mu : T \to \mathbb{R}$ and $k : T \times T \to \mathbb{R}$ which describe the first two moments:

Mean Function: $\mu(t) = \mathbb{E}[X_t]$

Covariance Function: $k(t, t') = \mathrm{Cov}(X_t, X_{t'})$

Gaussian Process Regression

Gaussian Processes are commonly used as a model for estimating an unknown function $f : T \to \mathbb{R}$ based on observed data.
We place a Gaussian Process prior $f \sim \mathcal{GP}(\mu, k)$ on f and optimize via maximum likelihood. Properties such as smoothness can be enforced via the kernel $k(\cdot, \cdot)$.

Gaussian Process Latent Variable Models

Key Idea: Using Gaussian Process regression, we can look for latent variables with non-linear relationships with the observations.

GP-LVM Model: Each observation $\mathbf{y}_i$ is generated by latent variables $\mathbf{x}_i$ according some unknown function $g : \mathbb{R}^Q \to \mathbb{R}^D$.

$$\mathbf{y}_i = g(\mathbf{x}_i) + \boldsymbol{\epsilon}_i$$

Practical Simplification

- Typically, g is modelled as D unknown real-valued functions; $g_1(\mathbf{x}), \ldots, g_D(\mathbf{x})$.

Gaussian Process Latent Variable Models

Important Notes

- Instead of using a Gaussian Process regression to learn $\mathbb{E}[\mathbf{y}_i|\mathbf{x}_i]$, GP-LVM reverses this procedure!
- In GP-LVM, the Q-dimensional latent variable representations $\{\mathbf{x}_i\}_{i=1}^N$ are learned in tandem with the functions $g_1, \ldots, g_D$.

Gaussian Process Latent Variable Models

Benefits of GP-LVM

- Very flexible application to non-linear functions.
- Admits a probabilistic interpretation.

Downsides of GP-LVM

- Highly expensive to compute, and requires iterative optimization algorithms.
- Does not admit a direct mapping from $\mathbb{R}^D$ to the latent space.

Fashion-MNIST Dataset

We use the Fashion-MNIST dataset (Xiao et al., 2017) as our source of image data.

The dataset contains 70,000 images of fashion items (clothes and accessories) which are classified in one of ten groups. By default, Fashion-MNIST is split into batches of 60,000 training images and 10,000 test images.

Each image is a 28×28 greyscale image, represented mathematically by an observation $\mathbf{y}_i \in \mathbb{R}^{784}$.

Fashion-MNIST Examples

Figure: Examples of each fashion item in the Fashion-MNIST dataset.

Preliminary Analysis

First, we create Scree plots for the principal components of the `Fashion-MNIST` training dataset $\mathbf{Y}^{\text{train}} \in \mathbb{R}^{60000 \times 784}$.

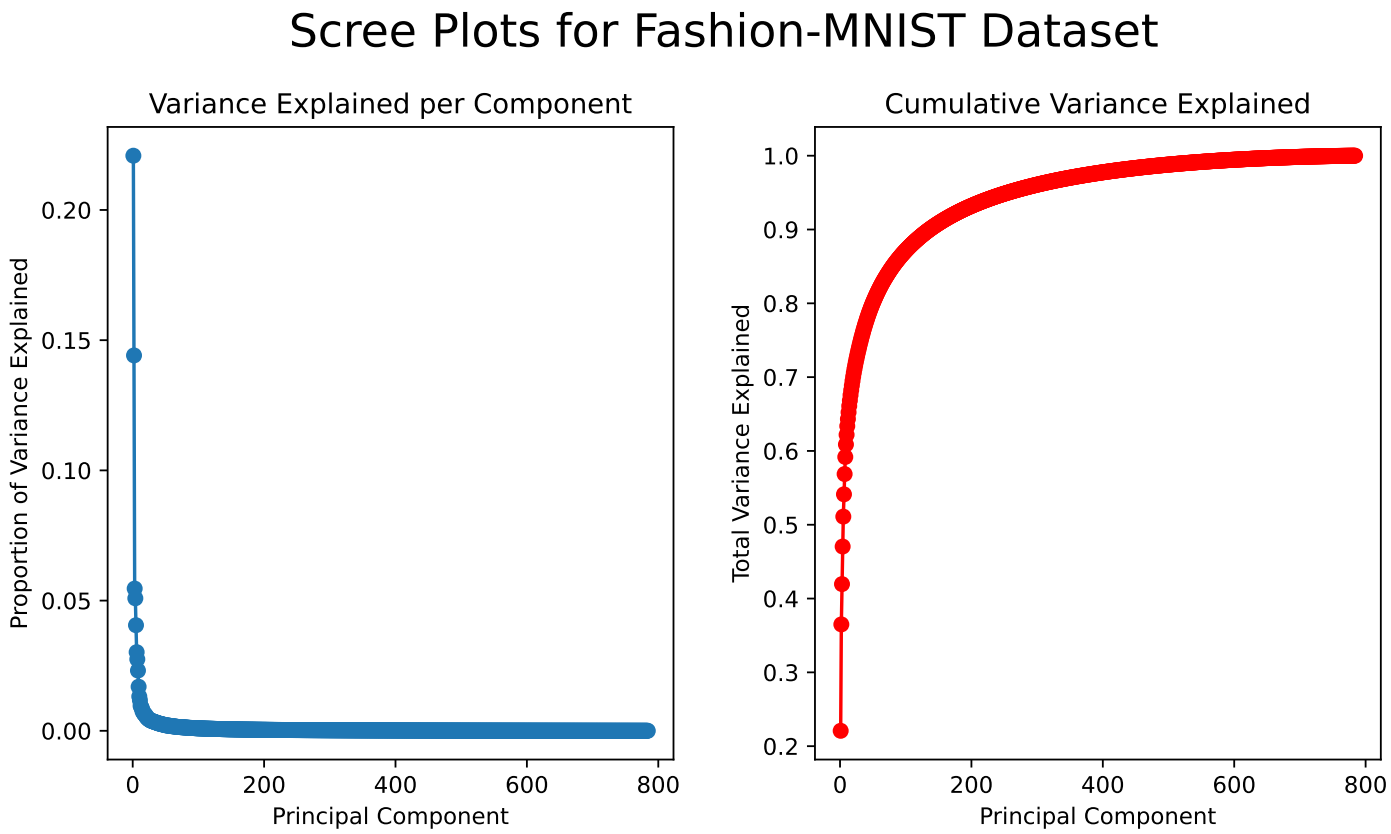


Figure: Scree plots for the principal components of the training data in the `Fashion-MNIST` dataset

Selection of Q

The Scree plots do not depict strong 'elbows' where we can clearly delineate important variables.

Cumulative Variance Explained

- The first 49 components explain 80% of the variance in the data.
- The first 136 components explain 90% of the variance in the data.

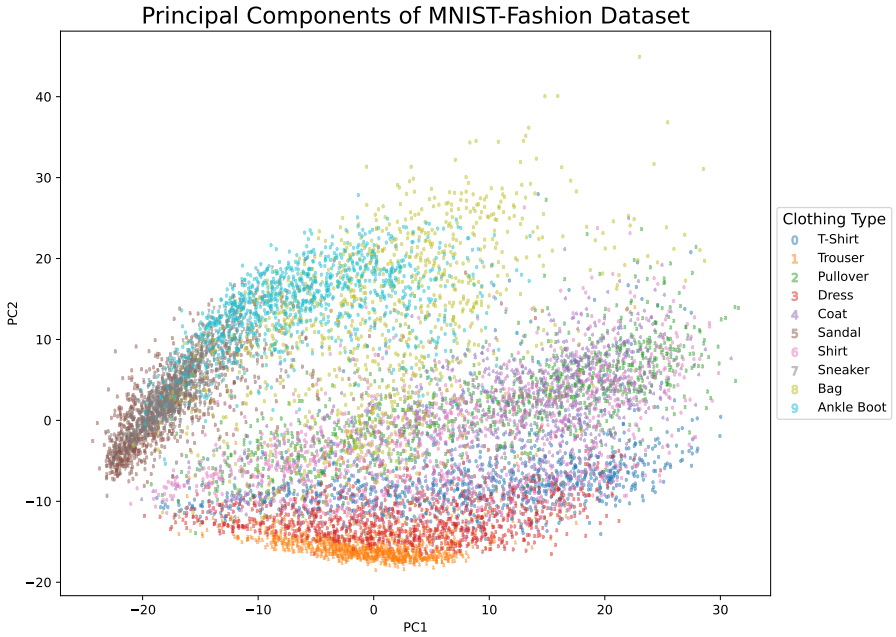
Final Selection of Q

- For visualizations and interpretations, we use $Q = 2$ and $Q = 3$.
- For creating our models, we use $Q = 10$.

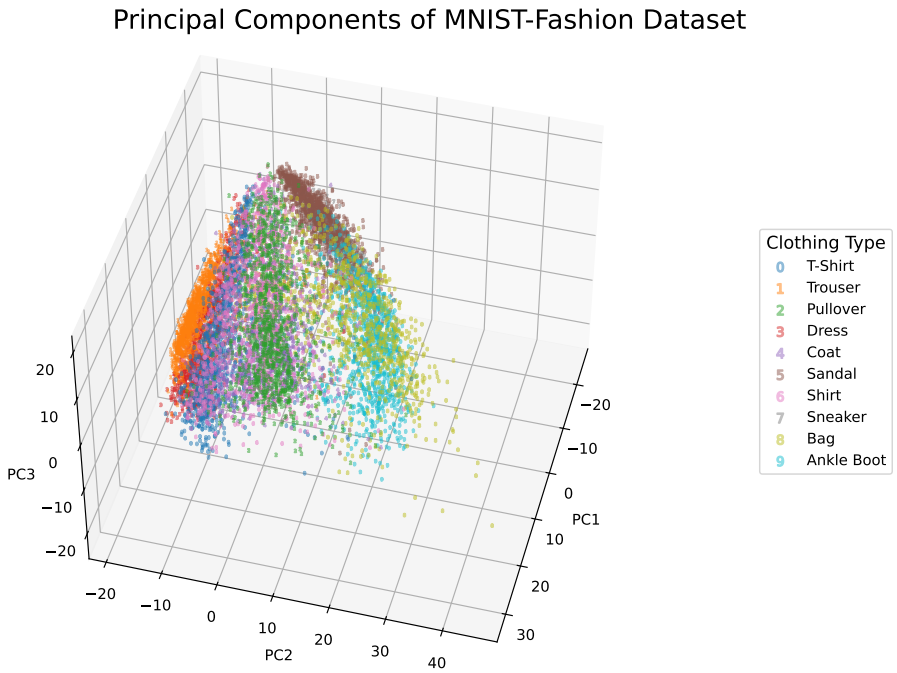
Model Comparison

To reduce the overall computational load, we reduce the dataset $\mathbf{Y}^{\text{train}}$ from 60,000 observations to 10,000 observations.
We stratify the sub-sampled data in order to keep the proportions of labels the same in the reduced dataset.
For all three model types (PCA, PPCA, GP-LVM), we reduce the sub-sampled data to $Q = 10$ dimensions, and use K-means to cluster data in the reduced space.

PCA Results



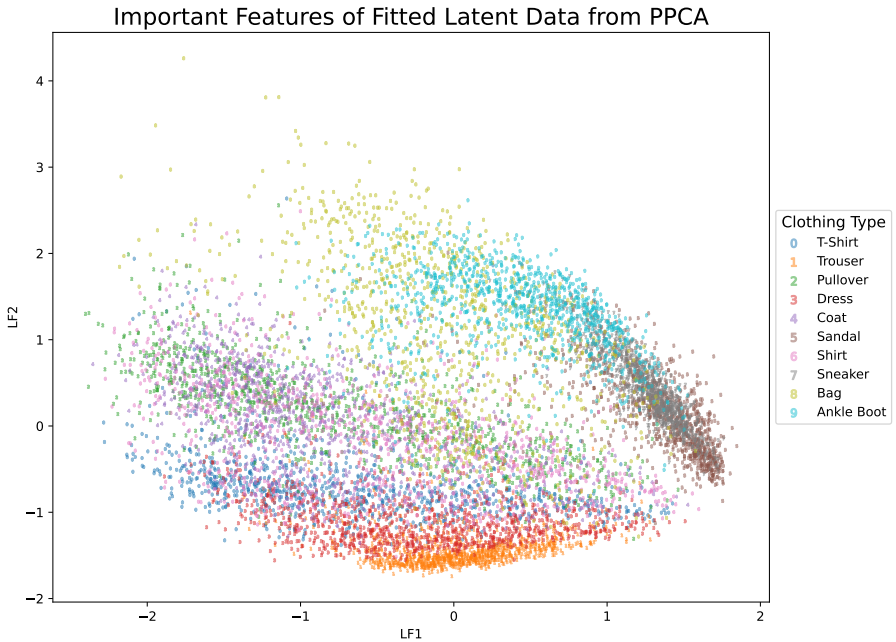
(a) Top two principal components of clothing items in the `Fashion-MNIST` dataset



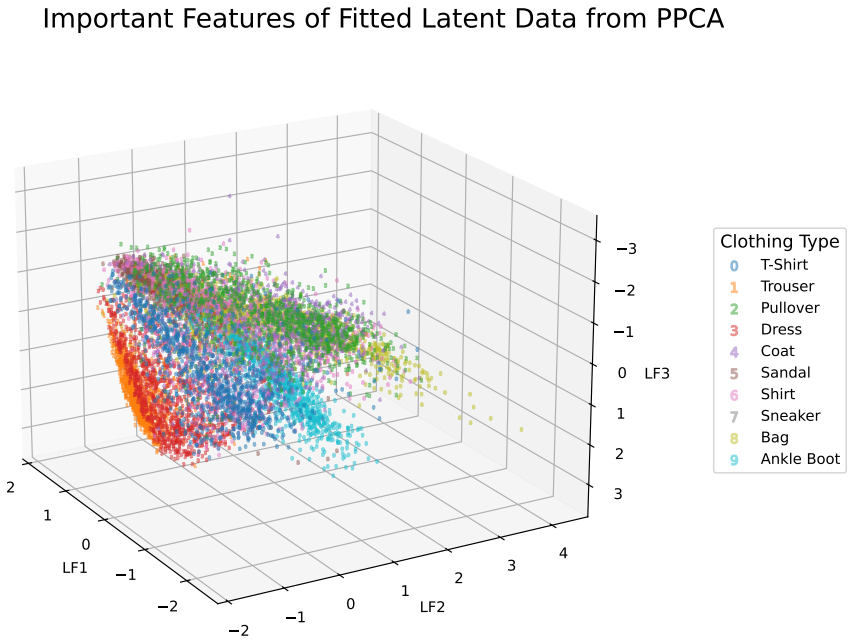
(b) Top three principal components of clothing items in the `Fashion-MNIST` dataset

Figure: Top principal components of clothing items, coloured by clothing type.

PPCA Results



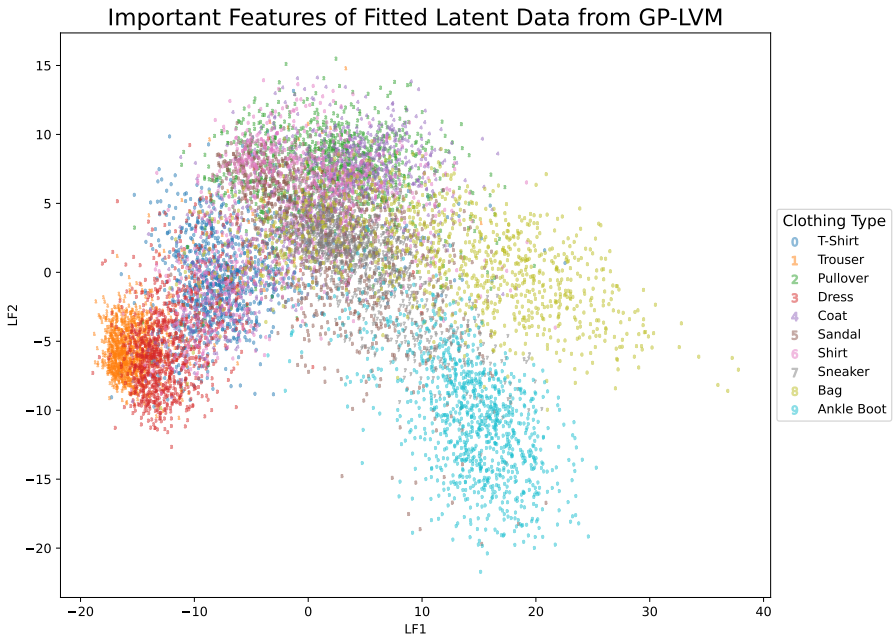
(a) Top two latent variables from PPCA for the `Fashion-MNIST` data



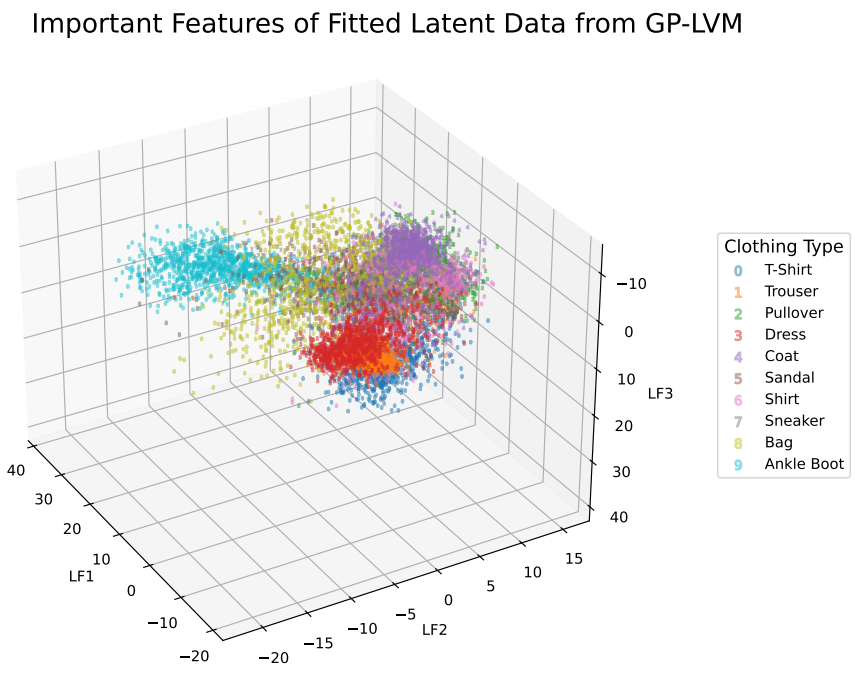
(b) Top three latent variables from PPCA for the `Fashion-MNIST` data

Figure: Top PPCA latent variables for representing images of clothing items, coloured by clothing type.

GP-LVM Results



(a) Top two latent variables from GP-LVM for the `Fashion-MNIST` data



(b) Top three latent variables from GP-LVM for the `Fashion-MNIST` data

Figure: Highest-importance GP-LVM latent variables for representing images of clothing items, coloured by clothing type.

Final Comparison

Method	Adjusted Rand Index	Reconstruction RMSE
PCA	35.06%	N/A
PPCA	34.63%	0.6114
GPLVM	38.06%	0.0000

Table: Dimensionality Reduction Results with $Q = 10$.

Limitations

Computational Costs

- Inferring the latent matrix $\mathbf{X} \in \mathbb{R}^{N \times Q}$ is *incredibly* expensive, limiting scalability.
- We had to reduce the size of both N and Q to perform the analysis, and we were unable to perform out-of-sample prediction for the GP-LVM model.

Further Research

There are several avenues for future research to both overcome the mentioned limitations and to further scale this project.

Future Work

- Working with different values of Q (particularly larger Q) to assess differences in performance.
- Incorporate out-of-sample predictions using $\mathbf{Y}^{\text{test}}$.
- Utilize more refined GP-LVM methods for improved performance.

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